

Introduction to Linear Regression

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Problem

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Assume that the production y of the i - th firm is described by

$$y = f(k, l) \quad (1)$$

where (k) is capital and (l) is labor.

Production

- If we want to duplicate the production is it enough to simply duplicate the input factors?
- How many unities of labor (l) must be reduced if increase the uses of one unity of capital (k)

Production function

Functional form

$$\ln(y) = \ln(A) + \alpha \ln(k) + \beta \ln(l) \quad (2)$$

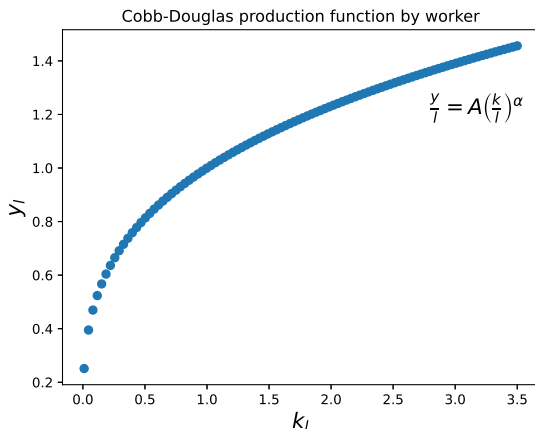
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- if $\alpha + \beta = 1$ the answer is yes in otherwise not.
- the marginal rate of substitution is $\frac{dk}{dl} = \frac{k}{l} \frac{\alpha}{\beta}$.

process data

We only need transform our variables in data-set in a logarithmic form and fit our model.

Cobb-Douglas by worker



Production by worker

$$\ln(y_L) = \ln(A) + \alpha \ln(k_L) \quad (3)$$

Two approaches

We are going to discuss two approaches; OLS and Maximum Likelihood Estimation (MLE) the first give us a intuitive idea, and the second give us a set of assumptions to inferential statistical purposes.

A linear model

Assuming that our variable is described by:

$$y_i = \beta_0 + \beta_1 x_i + u_i \quad (4)$$

Where are going consider that our prediction $\hat{y}_i$ differ from the real value y_i in u_i :

$$y_i - \hat{y}_i = u_i \quad (5)$$

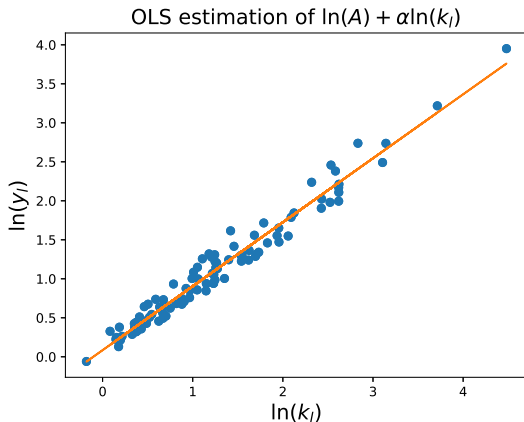
thus:

$$\hat{y}_i = \beta_0 + \beta_1 x_i \quad (6)$$

Mathematical problem

$$\min \sum (y_i - \hat{y}_i)^2 = \min \sum u_i^2 \quad (7)$$

Ordinary Least Squared (OLS)



Minimization problem

$$\min_{\beta_1, \beta_0} \sum (y_i - \hat{y}_i(\beta_0, \beta_1))^2 \quad (8)$$

Chain rule

$$\frac{\partial (\sum u_i^2)}{\partial \beta_k} = \sum \frac{\partial u_i^2}{\partial u_i} \frac{\partial u_i}{\partial \beta_k} = 0 \quad k = 0, 1. \quad (9)$$

Remember that $\frac{d \sum g_i(x)}{dx} = \sum \frac{dg_i(x)}{dx}$. Therefore $\frac{\partial u^2}{\partial u} = 2u$ and $\frac{\partial u}{\partial \beta_0} = -1$, $\frac{\partial u}{\partial \beta_1} = -x_i$.

$$\begin{aligned} \frac{\partial u_i}{\partial \beta_0} &= -2 \sum u_i = 0 \\ \frac{\partial u_i}{\partial \beta_1} &= -1 \sum u_i x_i = 0 \end{aligned}$$

rewriting the equations:

$$\frac{\partial u_i}{\partial \beta_0} = \sum (y_i - \beta_0 - \beta_1 x_i) = 0$$
$$\frac{\partial u_i}{\partial \beta_1} = \sum (y_i - \beta_0 - \beta_1 x_i) x_i = 0$$

We solve the First Order Conditions.

Solving Analytically

$$\hat{\beta}_1 = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = \frac{\text{cov}(x, y)}{\text{var}(x)}. \quad (10)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad (11)$$

See appendix to view the derivation of the formulas.

MLE

We have a collection of $\vec{y} = y_1, y_2, \dots, y_n$, that our objective is build a model that allow us to find the value of β_0, β_1 and maximize the probability of seen the data($\vec{y}$).

Explanation of MLE and card problems

$$\max_{\beta_0, \beta_1} \prod f(y_i | x_i, \beta_0, \beta_1, \sigma^2) \quad (12)$$

We already specify that $y_i = \beta_0 + \beta_1 x_i + u_i$

$$u_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \quad (13)$$

Therefore the $E(y_i) = \beta_0 + \beta_1 x_i$ and $\text{var}(y_i) = \sigma^2$ (why?).

$$y_i \stackrel{\text{iid}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2) \quad (14)$$

If we make distributional assumptions we could make inference and hypothesis testing.

PMF normal distributon

The PMF of a normal variable is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right) \quad (15)$$

where μ is mean. Therefore for our problem.

$$f(y_i) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{y_i - \beta_0 - \beta_1 x_i}{\sigma}\right)^2\right) \quad (16)$$

Likelihood function

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n f(y_i) \quad (17)$$

$$\prod_{i=1}^n f(y_i) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{y_i - \beta_0 - \beta_1 x_i}{\sigma}\right)^2\right) \quad (18)$$

the *likelihood function* is rewritten as:

$$\frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left(\frac{-1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right) \quad (19)$$

Log likelihood function

Apply natural logarithm to the likelihood function

$$\ln(L(\beta_0, \beta_1, \sigma^2)) = -n \ln(\sigma \sqrt{2\pi}) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \quad (20)$$

now we need establish the partial derivatives for β_0 and β_1 . Notice that to maximize the expression we need minimize $\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$.

$$\frac{\partial \ln(L)}{\partial \beta_0} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$
$$\frac{\partial \ln(L)}{\partial \beta_1} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i = 0$$

We are going to reach the same equations minimizing the error among the prediction and the real value of y_i .

$\hat{\beta}_1$ unbiased

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \quad (21)$$

Now we want probe that $\mathbb{E}[\hat{\beta}_1] = \beta_1$.

We can rewrite:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})y_i - \bar{y}\sum (x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \quad (22)$$

$$\begin{aligned}
 \hat{\beta}_1 &= \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2} \\
 &= \frac{\sum (x_i - \bar{x}) (\beta_0 + \beta_1 x_i + \varepsilon_i)}{\sum (x_i - \bar{x})^2} \\
 &= \frac{\left(\beta_0 \sum (x_i - \bar{x}) + \beta_1 \sum (x_i - \bar{x}) x_i + \sum (x_i - \bar{x}) \varepsilon_i \right)}{\sum (x_i - \bar{x})^2}
 \end{aligned}$$

$$\mathbb{E}[\hat{\beta}] = \mathbb{E} \left[\frac{\beta_1 \sum (x_i - \bar{x}) x_i + \sum (x_i - \bar{x}) \varepsilon_i}{\sum (x_i - \bar{x})^2} \right] \quad (23)$$

..

$$\mathbb{E} \left[\beta_1 + \frac{\sum (x_i - \bar{x}) \varepsilon_i}{\sum (x_i - \bar{x})^2} \right] \quad (24)$$

Unbiased estimator

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 \quad (25)$$

Variance

According to equation 17

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2} \quad (26)$$

Uses in the appendix and assume that the residuals are variables are independent.

$$\text{var}(\hat{\beta}_1) = \left(\frac{1}{\sum (x_i - \bar{x})^2} \right)^2 \sum (x_i - \bar{x})^2 \text{var}(y_i) \quad (27)$$

Here there are some important points remember that randomness come from ε .

$$\text{var}(y_i) = \sigma^2 \quad \forall i, i = 1, \dots, N. \quad (28)$$

$$\text{var}(\beta_1) = \left(\frac{1}{\sum (x_i - \bar{x})^2} \right)^2 \sigma^2 \sum (x_i - \bar{x})^2 \quad (29)$$

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} \quad (30)$$

Simulation

...

(Click here)

hint

In the simulation the approach of create a fixed population and sampling underestimated the variability because of give us the variability over these fixed set of errors, therefore we create errors in each trial (simulating more precisely different errors collecting new data).

Unbiasedness in Matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (31)$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}\mathbf{X}^T)^{-1} \mathbf{X}^T \mathbf{y} \quad (32)$$

$$\boldsymbol{\beta} = (\mathbf{X}\mathbf{X}^T)^{-1} \mathbf{X}^T (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \quad (33)$$

...

$$\hat{\beta} = (\mathbf{X}\mathbf{X}^T)^{-1} \mathbf{X}^T \mathbf{X} \beta + (\mathbf{X}\mathbf{X}^T)^{-1} \mathbf{X}^T \varepsilon \quad (34)$$

$$\hat{\beta} = \mathbf{I} \beta + (\mathbf{X}\mathbf{X}^T)^{-1} \mathbf{X}^T \varepsilon \quad (35)$$

Variance OLS

remember that

$$y = X\beta + \mu \quad \mu \sim N(0, \sigma^2 I) \quad (36)$$

now $\text{var}(y) = \sigma^2 I$.

$$\begin{aligned} \text{var}(\hat{\beta}) &= \text{var}((X^T X)^{-1} X^T y) \\ &= (X^T X)^{-1} X^T \text{var}(y) X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1} \end{aligned}$$

Simulation

...

(Click here)

hint

In the simulation the approach of create a fixed population and sampling underestimated the variability because of give us the variability over these fixed set of errors, therefore we create errors in each trial (simulating more precisely different errors collecting new data).

Hypothesis testing

Now we have the following information:

Expectation

$$\mathbb{E}[\hat{\beta}] = \beta \quad (37)$$

Variance

$$\text{var}(\hat{\beta}) = \sigma^2 \left(\mathbf{X}^T \mathbf{X} \right)^{-1} \quad (38)$$

Summary properties

$$\sum_{i=1}^n x_i = x_1 + x_2 + x_3 + \dots + x_n \quad (39)$$

$$\sum a = na \quad (40)$$

$$\sum(ax) = a \sum x \quad (41)$$

$$\sum(x_i + y_i) = \sum x_i + \sum y_i \quad (42)$$

$$\sum(ax + by) = a \sum x_1 + b \sum y_1 \quad (43)$$

$$\begin{aligned}\sum(x_i - \bar{x}) &= 0 \\ n\bar{x} - n\bar{x} &= 0\end{aligned}$$

$$\begin{aligned} & \sum x_i y_i - n \bar{x} \bar{y} \\ &= \sum x_i y_i - \sum x_i \bar{y} \\ &= x_1 y_1 + x_2 y_2 + \dots + x_n y_n - (x_1 + x_2 + \dots + x_n) \bar{y} \\ &= x_1 y_1 - x_1 \bar{y} + x_2 y_2 - x_2 \bar{y} + \dots + x_n y_n - x_n \bar{y} \\ &= x_1 (y_1 - \bar{y}) + x_2 (y_2 - \bar{y}) + \dots + x_n (y_n - \bar{y}) \\ &= \sum x_i (y_i - \bar{y}) = \sum y_i (x_i - \bar{x}) \end{aligned}$$

$$\begin{aligned}\sum (x_i - \bar{x})^2 &= \sum (x_i - \bar{x})(x_i - \bar{x}) \\ &= \sum ((x_i - \bar{x})x_i - (x_i - \bar{x})\bar{x}) \\ &= \sum (x_i - \bar{x})x_i - \bar{x} \sum (x_i - \bar{x})^0\end{aligned}$$

$$\sum (x_i - \bar{x})^2 = \sum (x_i - \bar{x})x_i \quad (44)$$

With this properties and examples we could verify that :

$$\sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 \quad (45)$$

$$\sum (y_i - \bar{y})(x_i - \bar{x}) = \sum x_i y_i - n\bar{x}\bar{y} \quad (46)$$

Solving FOC

rewriting the equations:

$$\begin{aligned}\sum (y_i - \beta_0 - \beta_1 x_i) &= 0 \\ \sum (y_i - \beta_0 - \beta_1 x_i) x_i &= 0\end{aligned}$$

applying:

$$\begin{aligned}n\bar{y} - n\beta_0 - n\beta_1\bar{x} &= 0 \\ \sum x_i y_i - n\beta_0\bar{x} - \beta_1 \sum x_i^2 &= 0\end{aligned}$$

Solving FOC

now of $n\bar{y} - n\beta_0 - n\beta_1\bar{x}$ we can get:

$$\beta_0 = \bar{y} - \beta_1\bar{x} \quad (47)$$

now replace β_0 in $\sum x_i y_i - n\beta_0\bar{x} - \beta_1 \sum x_i^2$

$$\begin{aligned} \sum x_i y_i - n\bar{x}\bar{y} + n\beta_1\bar{x}^2 - \beta_1 \sum x_i^2 &= 0 \\ \sum x_i y_i - \bar{x}\bar{y} &= \beta_1(\sum x_i^2 - n\bar{x}^2) \\ \beta_1 &= \frac{\sum x_i y_i - \bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} \end{aligned}$$

According to the properties defined previously:

$$\beta_1 = \frac{\sum x_i y_i - \bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = \frac{\text{cov}(x, y)}{\text{var}(x)}. \quad (48)$$

Two dice sample space and PMF

6	7	8	9	10	11	12
5	6	7	8	9	10	11
4	5	6	7	8	9	10
3	4	5	6	7	8	9
2	3	4	5	6	7	8
1	2	3	4	5	6	7
	1	2	3	4	5	6

x	$f(x)$
2	$\frac{1}{36}$
3	$\frac{2}{36}$
4	$\frac{3}{36}$
5	$\frac{4}{36}$
6	$\frac{5}{36}$
7	$\frac{6}{36}$
8	$\frac{5}{36}$
9	$\frac{4}{36}$
10	$\frac{3}{36}$
11	$\frac{2}{36}$
12	$\frac{1}{36}$

Expected value

Die game

Expected value (Lab)

Discrete definition

$$\mu_X = \mu = E(X) = \sum_x f(x)x \quad (49)$$

Continuous definition

$$E(x) = \int_{-\infty}^{\infty} xf(x) \quad (50)$$

Join distribution

until now we try Ω in $\mathbf{R}^1$ and we can be interested in find the probability of occurrence of two simultaneous random variables.

$$f(x, y) = P(X = x, Y = y) \quad (51)$$

Some intuitive properties are:

- $f(x, y) \geq 0$
- $\sum_x \sum_y f(x, y) = 1$

Excercise

Suppose the bag model with n balls and there there are r balls and w balls where $r + w = n$ find the probability of get x, y balls respectively.

Marginal distribution

From the joint distribution $f(x, y) = P((X = x) \cap P((Y = y))$

$$g(x) = \sum_y f(x, y) \quad (52)$$

$$h(y) = \sum_x f(x, y) \quad (53)$$

Joint Probability Function

		X				
		x_1	x_2	$\dots$	x_m	$f(y)$
Y	y_1	$f(x_1, y_1)$	$f(x_2, y_1)$	$\dots$	$f(x_m, y_1)$	$f(y = 1) = \sum_{i=1}^m f(x_i, y_1)$
	y_2	$f(x_1, y_2)$	$f(x_2, y_2)$	$\dots$	$f(x_m, y_2)$	$f(y = 2) = \sum_{i=1}^m f(x_i, y_2)$
	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
	y_n	$f(x_1, y_n)$	$f(x_2, y_n)$	$\dots$	$f(x_m, y_n)$	$f(y = n) = \sum_{i=1}^m f(x_i, y_n)$
	$f(x)$	$f(x = 1) = \sum_{i=1}^n f(x_1, y_i)$	$f(x = 2) = \sum_{i=1}^n f(x_2, y_i)$	$\dots$	$f(x = m) = \sum_{i=1}^n f(x_m, y_i)$	

Expected value

Property

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \quad (54)$$

remember that $P(x, y) = P((X = x) \cap (Y = y))$, and $P(x) = \sum_y P(x, y)$ and $P(y) = \sum_x P(x, y)$

...

$$\mathbb{E}[X + Y] = \sum (x + y)P(x, y) \quad (55)$$

consider that $\sum_{x,y} = \sum_x \sum_y$

...

$$\mathbb{E}[X + Y] = \sum_{x,y} (x)P(x, y) + \sum_{x,y} (y)P(x, y)$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \quad (56)$$

$$\begin{aligned}\mathbb{E}[XY] &= \sum_{x,y} xyP(x,y) \\ &= \sum_{x,y} xyP(x)p(y) \\ &= \sum_{x,y} xP(x)yP(y) \\ &= \sum_x xp(x) \sum_y yP(y)\end{aligned}$$

Property

proof with integrals that

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b \quad (57)$$

Excercise

Simulate and check by definition what is the average profit if you pay to a game the amount the rolled number if a throwing of a dice.

Variance

$$\begin{aligned}\sigma^2 &= E[(X - \mu)^2] \\ &= E(x^2) - [E(X)]^2\end{aligned}$$

remember that $\sum_x f(x)x = \mu$ and $\sum_x f(x) = 1$

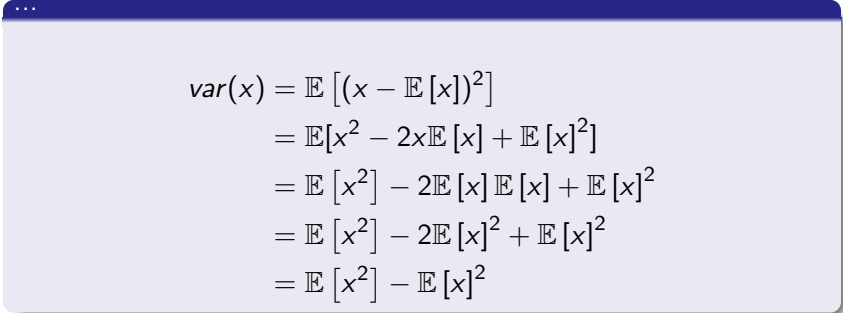
$$\begin{aligned}\sigma^2 &= \sum_x (x - \mu)^2 f(x) \\ &= \sum_x (x^2 f(x) - 2x\mu f(x) + \mu^2 f(x)) \\ &= \sum_x x^2 f(x) - \mu^2\end{aligned}$$

Variance

The following are important properties about variance:

- $\text{var}(x) \geq 0$
- $\text{var}(c) = 0$
- $\text{var}(cx) = c^2 \text{var}(x)$
- $\text{var}(x + c) = \text{var}(x)$
- $\text{var}(x) = \mathbb{E}(x^2) - \mathbb{E}^2(X)$
- $\text{Var}(x + y) \neq \text{var}(x) + \text{var}(y)$

Property

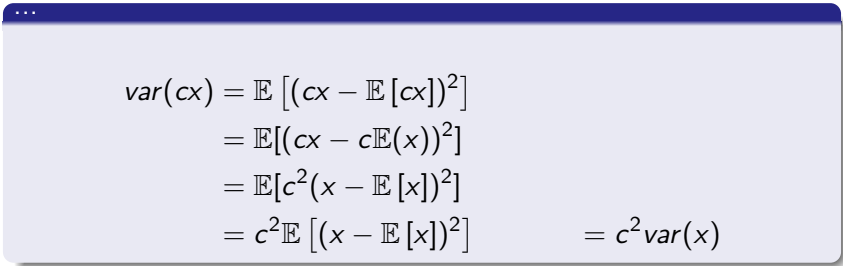

$$\begin{aligned}\text{var}(x) &= \mathbb{E}[(x - \mathbb{E}[x])^2] \\ &= \mathbb{E}[x^2 - 2x\mathbb{E}[x] + \mathbb{E}[x]^2] \\ &= \mathbb{E}[x^2] - 2\mathbb{E}[x]\mathbb{E}[x] + \mathbb{E}[x]^2 \\ &= \mathbb{E}[x^2] - 2\mathbb{E}[x]^2 + \mathbb{E}[x]^2 \\ &= \mathbb{E}[x^2] - \mathbb{E}[x]^2\end{aligned}$$

Property

$$\begin{aligned}\text{var}(c) &= \mathbb{E} [(c - \mathbb{E}[c])^2] \\ &= \mathbb{E} [(c - c)^2] = 0.\end{aligned}$$

Note remember that $\mathbb{E}[c] = c$.

Property


$$\begin{aligned} \text{var}(cx) &= \mathbb{E} [(cx - \mathbb{E}[cx])^2] \\ &= \mathbb{E} [(cx - c\mathbb{E}(x))^2] \\ &= \mathbb{E} [c^2(x - \mathbb{E}[x])^2] \\ &= c^2 \mathbb{E} [(x - \mathbb{E}[x])^2] &= c^2 \text{var}(x) \end{aligned}$$

Property

Proof that $\text{var}(x + c) = \text{var}(x)$.

$$\begin{aligned}\text{var}(x + c) &= \mathbb{E} [((x + c) - \mathbb{E}[x + c])^2] \\ &= \mathbb{E} [(x + c - \mathbb{E}(x) - c)^2] \\ &= \mathbb{E} [(x - \mathbb{E}[x])^2] \\ &= \text{var}(x)\end{aligned}$$

Property

$$\begin{aligned} \text{var}(mX + nY) &= \mathbb{E}[(mX + nY - \mathbb{E}[mX + nY])^2] \\ &= \mathbb{E}[(m(X - \mathbb{E}[X]) + n(Y - \mathbb{E}[Y]))^2] \\ &= \mathbb{E}[m^2(X - \mu_X)^2] + \mathbb{E}[n^2(Y - \mu_Y)^2] \\ &\quad + \mathbb{E}[2mn(X - \mu_X)(Y - \mu_Y)] \\ &= m^2 \text{var}(X) + n^2 \text{var}(Y) + 2mncov(X, Y) \end{aligned}$$

if $cov(X, Y) = 0$ then:

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y). \quad (58)$$

Propertie

With all properties we can derive another important.

- $var(mX + b) = m^2 var(X)$

Insights

Think in θ as a unknown parameter, but in **Classical statistics** this parameters is **fixed** our work is find it.

We need a **estimator**, that process the data observed, in other words a **estimator** is a function ($\hat{\theta}$) aiming to find θ . In the process $\hat{\theta}(x)$ is a estimate of a particular set(x) of X .

Think in variance that we have two possible estimators(which are?), and $\theta(\hat{x})$ could be a specific value of them.

Estimation

Now we are interested in find a **estimator** $\hat{\Theta}$ that

$$\min(\hat{\theta} - \theta) \quad (59)$$

Namely, we are finding that the estimated value of the unknow parameter is small in relation with the true value.

Estimation of mean

A estimator of mean is $\hat{\Theta} = \frac{\sum x_i}{n}$. note that $\hat{\Theta}$ is random variable.
we are interested that a estimator have the following properties;

Unbiased and **consistency**.

- $E(\hat{\Theta}) = \theta$
- By Weak Law Larger Numbers $\hat{\Theta} \rightarrow \theta$ when $n \rightarrow \infty$ (more data, higher accuracy in the estimation).

The error of a estimator

Remember that $\hat{\theta}$ is a random variable, Therefore Mean squared error(MSE) $E[(\hat{\theta} - \theta)^2]$, could be expressed as

$$E[(\hat{\theta} - \theta)^2] = \text{var}(\hat{\theta} - \theta) + (E[(\hat{\theta} - \theta)])^2 \quad (60)$$

Applying properties of variance (variance plus or minus a constant is the variance of the r.v).

$$\text{var}(\hat{\theta}) + (E[(\hat{\theta} - \theta)])^2 \quad (61)$$

$$MSE = E[(\hat{\theta} - \theta)^2] = \text{var}(\hat{\theta}) + \text{bias}^2 \quad (62)$$

Analysis about mean estimator

We known that the

$$MSE(\hat{\theta}) = \frac{\sigma^2}{n} + 0 \quad (63)$$

According to the consistency property then the error is lesser wick the size of data increase $n \rightarrow \infty$.

it is important note here that if bias is equal to zero, then we avoid that the error rely on about θ .

standard error SE .

Is a important concept in estimator

$$SE = \sqrt{\text{var}(\hat{\Theta})} \quad (64)$$

Property

Assume that $X \stackrel{iid}{\sim} (\mu, \sigma^2)$ then:

$$\begin{aligned} \text{var}(\bar{x}) &= \text{var}\left(\frac{\sum x_i}{n}\right) \\ &= \frac{1}{n^2} \sum \text{var}(x_i) \\ &= \frac{1}{n^2} n\sigma^2 \\ &= \frac{\sigma^2}{n} \end{aligned}$$

$$\sqrt{\text{var}(x)} = \text{sdv}(\bar{x}) = \frac{\sigma}{\sqrt{n}} \quad (65)$$

Degree of freedom

For instance if said that $\bar{x} = 10$, and if we take for values 5 for instance:

$$3 - 10 = -7 \quad (66)$$

$$10 - 10 = 0 \quad (67)$$

$$14 - 10 = 4 \quad (68)$$

$$11 - 10 = 1 \quad (69)$$

$$\mathbf{x} - 10 = \mathbf{2} \quad (70)$$

Notice, that the first four number could be any value, however the last value is attached to fill the propertie of sum equal to zero, therefore $\mathbf{x}$ must be equal to 12.

How many degree of fredoms we have?

μ and σ^2 estimators

$\bar{X}$ is a estimator of μ and S^2 is a estimator of population variance σ^2 .

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n} \quad (71)$$

$$S^2 = \frac{\sum (x_i - \mu)^2}{n - 1} \quad (72)$$

Why divide in S^2 by $n - 1$?

Variance estimators

We can check mathematically (**See here**) that S_N^2 is a biased estimator

$$S_n^2 = \frac{\sum (x_i - \bar{x})^2}{n} \quad (73)$$

and that S_{n-1}^2 is unbiased.

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad (74)$$

However to see a simulation **click here**

Relate

$$\begin{aligned} \text{var}(x) &= \sigma^2 \\ &= \mathbb{E}[(x - \mu_x)]^2 \\ &= \mathbb{E}[x^2] - 2\mu_x^2 \\ &= \mathbb{E}[x^2] - \mu_x^2 \end{aligned}$$

$$\mathbb{E}[x^2] = \sigma^2 + \mu_x^2 \quad (75)$$

$$E(s^2) = \sigma^2$$

proof

To make this proof we need related the sample variance with the population variance,

$$E(x^2) = \sigma^2 + \mu^2$$

remember that $\text{var}(\bar{x}) = \frac{\sigma^2}{n}$

$$E(\bar{x}^2) = \frac{\sigma^2}{n} + \mu^2$$

now aplying the expected value definition (and properties) to variance sampling estimator we have:

$$E[s^2] = \frac{1}{n-1} (nE(x_i^2) - nE(\bar{x}^2)) \quad (76)$$

reeplacing and applyinng basic algebra operations then $E(s^2) = \sigma^2$.

Theorem 1

X is a random variable with a pdf $f(x)$ then μ of $g(x)$ is

$$\mu_{g(x)} = E(g(x)) = \sum g(x)f(x) \quad (77)$$

Example of income and the probability of sell a product.

Theorem

X is a random variable with pdf $f(x)$ then the variance of $g(x)$ will be:

$$\sigma_{g(x)}^2 = E((g(x) - \mu_{g(x)})^2) \quad (78)$$

this equation is derived from the definition of variance of a random variable, remember that $g(x)$ is a random variable with mean $\mu_{g(x)}$.

Expected value of two random variables

let be X, Y two random variables with joint probability function distribution $f(x, y)$ the mean of $g(X, Y)$ is:

$$\mu_{g(X,Y)} = \mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y) f(x, y) \quad (79)$$

Covariance

Insights

$$\begin{aligned}\sigma_{X,Y} &= E[(X - \mu_x)(Y - \mu_y)] \\ &= \sum_x \sum_y (x - \mu_x)(y - \mu_y)f(x, y)\end{aligned}$$

is a measure of association between two variables

not divide by $n - 1$

Here we are seeing the variance of the estimator.

$$\begin{aligned} E(\hat{\Theta}) &= E(\bar{x}) = E\left(\frac{\sum (x_i - \bar{x})^2}{n}\right) \\ &= \frac{1}{n} E\left(\sum x_i^2 - n\bar{x}^2\right) \end{aligned}$$

Applying some algebraic operations we have, remember as compute $E(x_i^2)$ and $E(\bar{x}^2)$.

$$E(\hat{\Theta}) = \frac{\sigma^2(n-1)}{n} \tag{80}$$

Appendix

Expectation

...

Assuming $\mathbf{x}$ a random variable:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad (81)$$

...

$$\mathbb{E}[\mathbf{x}] = \begin{bmatrix} \mathbb{E}[x_1] \\ \mathbb{E}[x_2] \\ \vdots \\ \mathbb{E}[x_n] \end{bmatrix} \quad (82)$$

$$\mathbb{E}[\mathbf{x} + \mathbf{x}'] = \begin{bmatrix} \mathbb{E}[x_1 + x'_1] \\ \mathbb{E}[x_2 + x'_2] \\ \vdots \\ \mathbb{E}[x_n + x'_n] \end{bmatrix} \quad (83)$$

Note that $\mathbb{E}[x_i + x'_i] = \mathbb{E}[x_i] + \mathbb{E}[x'_i]$ therefore:

$$\mathbb{E}[\mathbf{x}] + \mathbb{E}[\mathbf{x}'] \quad (84)$$

$$\mathbb{E}[\mathbf{x} + \mathbf{x}'] = \mathbb{E}[\mathbf{x}] + \mathbb{E}[\mathbf{x}'] \quad (85)$$

Expectation

...

A a matrix and **x** a random variable then:

$$\mathbb{E}[\mathbf{Ax}] = A\mathbb{E}[x] \quad (86)$$

Is expectation a **linear operator**. this property es extended to **X**.

Variance

$$\text{var}(x) = \mathbb{E} \left[(\mathbf{X} - \mu_x) (\mathbf{X} - \mu_x)^T \right] \quad (87)$$

$$\text{var}(x) = \mathbb{E} \left[\begin{pmatrix} x_1 - \mu_{x_1} \\ x_2 - \mu_{x_2} \\ \vdots \\ x_n - \mu_{x_n} \end{pmatrix} (x_1 - \mu_{x_1} \quad x_2 - \mu_{x_2} \quad \dots \quad x_n - \mu_{x_n}) \right]$$

Variance

...

The variance of a random vector $\mathbf{x}$ is a matrix (variance - covariance matrix) denominated $\mathbf{S}$ that is symmetric.

$$\text{var}(\mathbf{x}) = \mathbf{S} = [s_{ij}] = \begin{cases} \text{cov}(x_i, x_j) & \text{if } i \neq j, \\ \text{var}(x_i) & \text{if } i = j. \end{cases} \quad (88)$$

...

Variance for

$$\text{var}\left(\sum a_i x_i\right) = \sum a_i^2 \text{var}(x_i) + \sum_{1 \leq i < j \leq n} \text{cov}(x_i, x_j) \quad (89)$$

$$\begin{aligned}\text{var}(\mathbf{Ax}) &= \mathbb{E} \left[(\mathbf{Ax} - \mathbb{E}[\mathbf{Ax}]) (\mathbf{Ax} - \mathbb{E}[\mathbf{Ax}])^T \right] \\ &= \mathbb{E} \left[(\mathbf{Ax} - \mathbf{A}\mathbb{E}[x]) (x^T \mathbf{A}^T - \mathbb{E}[x]^T \mathbf{A}^T) \right] \\ &= \mathbb{E} \left[\mathbf{A}(x - \mathbb{E}[x]) (x - \mathbb{E}[x])^T \mathbf{A}^T \right] \\ &= \mathbf{A} \text{var}(x) \mathbf{A}^T\end{aligned}$$

This properties es extended to X .

Sum transpose

...
assume $C = [c_{ij}] = [a_{ij} + b_{ij}]$ $[c_{ij}]^T = [a_{ji} + b_{ji}] = [a_{ij}]^T + [b_{ij}]^T$

...
$$(A + B)^T = A^T + B^T \quad (90)$$

Transpose of inverse

$$\beta \mathbf{A} = \mathbf{I}$$

then $\mathbf{A} = \beta^{-1}$.

$$(\beta \mathbf{A})^T = \mathbf{I}^T$$

$$\mathbf{A}^T \beta^T = \mathbf{I}$$

finally we see that $\mathbf{A}^T = (\beta^T)^{-1}$ and therefore
 $(\beta^{-1})^T = (\beta^T)^{-1}$.

$$(\beta^{-1})^T = (\beta^T)^{-1} \quad (91)$$

$$\mathbf{y} = \beta \mathbf{X} + \epsilon \quad (92)$$

Assumptions

$$\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \quad \forall i, i = 1, \dots, N. \quad (93)$$

$$\varepsilon \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \sigma^2 \mathbf{I}) \quad (94)$$

where $\mathbf{I}$ is identity matrix.

The last statements summarize four assumptions.

- $\mathbb{E}[\varepsilon_i] = 0$
- $\text{var}(\varepsilon_i) = \sigma^2$
- $\text{cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \forall i \neq j$
- $\varepsilon_i \sim N(\mu, \sigma^2)$

Linearity variance

...

[linearity lab \(click here\)](#)

Research about assumption

- Linearity parameters.
- Random sampling
- Not perfect multicollinearity
- Conditional (to independent) error mean equal to zero (**exogeneity**).
- Homocedasticity
- No autocorrelation in error terms
- Residuals are distributed normally (check by the methods that you already know as Q-Q plot, or statistical test as Kolmogorov-smirnov test)

from where come beta?

$$J(\beta) = \|X\beta - y\|^2$$

$$\begin{aligned} J(\beta) &= (X\beta - y)^T (X\beta - y) \\ &= \beta^T X^T X \beta - \beta^T X^T y - y^T X \beta + y^T y \\ &= \beta^T X^T X \beta - 2\beta^T X^T y + y^T y \end{aligned} \quad (95)$$

$$\frac{\partial J(\beta)}{\partial \beta} = 2X^T X \beta - 2X^T y = 0 \quad (96)$$

$$X^T X \beta = X^T y \quad (97)$$

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad (98)$$